SLAB DESIGN CALCULATIONS

Project Name	:	G+2 RB SLAB DESIGN @FLOOR 2
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Slab-1-8.077 m.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 03:00:45

NOTES:

1. Slab is designed as per 1 M width of panel.
2. Density of concrete assumed as 25 kN/Cum.
3. Maximum spacing of Main steel is smaller of (3 x Deff and 300 mm)
4. Maximum spacing of Distribution steel is smaller of (5 x Deff and 450 mm)
5. All 6 mm diameter bars assumed of Fe250 grade with Fy=250 N/sq.mm

Definitions Of Terms:

1.	α	=	Coefficient of Thermal Expansion
2.	acr	=	Distance from the point considered to the surface of the nearest longitudinal bar
3.	εm	=	Average steel strain at level considered
4.	ε1	=	Strain at level considered, Calculated ignoring the stiffening of the concrete in tension zone
5.	ρact	=	Percentage steel provided
6.	ρcrit	=	Percentage steel critical as per code
7.	BM	=	Bending moment in kNm
8.	D	=	Overall depth of slab in mm
9.	Deff	=	Effective depth of slab in mm
10.	Ec	=	Modulus of elasticity of concrete
11.	Es	=	Modulus of elasticity of steel
12.	Fck	=	Characteristic compressive strength of concrete cube in N/sqmm
13.	Fst	=	Stress in steel
14.	Fy	=	Yield Stress Of Steel in N/sqmm
15.	Mu	=	Factored Bending moment in kNm
16.	Sp	=	Spacing Between bars at outer most layer
17.	Tc	=	Permissible Shear stress in concrete in N/sqmm
18.	Temp1	=	Peak Hydration temperature
19.	Temp2	=	Seasonal Temperature Variations
20.	Tv	=	Shear stress due to Vu in N/sqmm
21.	V	=	Shear force in kN
22.	Vu	=	Factored shear force in kN
23.	Wcr	=	Surface Crack Width

Code References :
IS 456

	ELEMENT		CLAUSE / TABLE
1.	Moment Coeffs For 2-Way Slabs	:	Greschoff-Rankine Formulation and yield line theory
2.	Ptmin	:	26.5.2.1
3.	Ptcalculated	:	38
4.	Shear Strength Of Concrete	:	40.3
5.	Modification Factor for Shear	:	40.2.1.1
6.	Max Nominal Shear Stress	:	40.2.3
7.	Basic Span To Depth Ratio	:	23.2.1
8.	Tensile Steel Factor	:	23.2.1
9.	Compression Steel Factor	:	23.2.1
10.	Flange correction factor	:	23.2.1
11.	spacing between main bars	:	26.3.3
12.	spacing between distribution bars	:	26.3.3
13.	crack width calculation	:	Annex-F
14.	cracking	:	Clause 35.3.2

BS 8007 (For Early / Initial Thermal Cracking)

	ELEMENT		CLAUSE / TABLE
1.	Surface Zone for suspended slab	:	Figure A.1
2.	Surface Zone for ground slab	:	Figure A.2
3.	Factors for the calculation of minimum reinforcement	:	Table A.1

Slab No. :	**S1**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.990	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm
Slab Thickness	=	150.000	mm

Effective Depth Along LX, Deffx	=	125.000	mm
Effective Depth Along LY, Deffy	=	115.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	2-Way	
Panel Type	=	Two Adjacent Edges Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.050	
BM	=	5.606	kNm
Mu = 1.5 X BM	=	8.410	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.067	
BM	=	7.475	kNm
Mu = 1.5 X BM	=	11.213	kNm
Area Of Reinforcement Required (Mu)	=	194.132	sqmm/m
Additional Area Of Reinforcement Required	=	104.132	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.034	
BM	=	3.788	kNm
Mu = 1.5 X BM	=	5.682	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.045	
BM	=	5.050	kNm
Mu = 1.5 X BM	=	7.576	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement	=	150.000	sqmm/m

Required			
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.492	
V = SF1 x TL x Lx	=	13.979	kN
Vu = 1.5 X V	=	20.969	kN
Shear Stress, Tv	=	0.168	N/sqmm
Nominal Shear Stress, Tc	=	0.296	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.384	
	>	0.168	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.400	
V = SF2 x TL x Lx	=	11.368	kN
Vu = 1.5 X V	=	17.052	kN
Shear Stress, Tv	=	0.148	N/sqmm
Nominal Shear Stress, Tc	=	0.297	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.386	
	>	0.148	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	36.380	
Lx/d	=	31.360	
	<	36.380	Slab Is Safe In Deflection

Slab No. :	**S2**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.990	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Effective Depth Along LY, Deffy	=	115.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	2-Way	
Panel Type	=	One Short Edge Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.038	
BM	=	4.252	kNm
Mu = 1.5 X BM	=	6.378	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.051	
BM	=	5.669	kNm
Mu = 1.5 X BM	=	8.504	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.028	
BM	=	3.064	kNm
Mu = 1.5 X BM	=	4.596	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.037	
BM	=	4.085	kNm
Mu = 1.5 X BM	=	6.127	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.435	
V = SF1 x TL x Lx	=	12.351	kN
Vu = 1.5 X V	=	18.527	kN
Shear Stress, Tv	=	0.148	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.148	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.360	
V = SF2 x TL x Lx	=	10.231	kN
Vu = 1.5 X V	=	15.347	kN
Shear Stress, Tv	=	0.133	N/sqmm
Nominal Shear Stress, Tc	=	0.297	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.386	
	>	0.133	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	36.380	
Lx/d	=	31.360	
	<	36.380	Slab Is Safe In Deflection

Slab No. :	**S3**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.990	m
Short Span, Lx	=	1.370	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Dead Load,DL	=	5.250	kN/sqm
Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	1-Way	
End Conditions	=	End Span	
Load Combination	=	1.5 DL + 1.5 LL	

Design Moments:

Short Span Positive Moment at Mid Span -

Moment Coefficient(DL), M1	=	0.083	
Moment Coefficient(LL), M2	=	0.100	
BM = M1 x DL x Lx x Lx + M2 x LL x Lx x Lx	=	1.197	kNm
Mu = 1.5 X BM	=	1.795	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Positive Reinforcement -

Area Of Reinforcement Required	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment at Continuous Support -

Moment Coefficient(DL), M3	=	0.100	
Moment Coefficient(LL), M4	=	0.111	
BM = M3 x DL x Lx x Lx + M4 x LL x Lx x Lx	=	1.402	kNm
Mu = 1.5 X BM	=	2.104	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

All Other Sections provide Minimum Steel @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Shear Coefficient(DL), SF1	=	0.400	
Shear Coefficient(LL), SF2	=	0.450	

V = SF1 x DL x Lx + SF2 x LL x Lx	=	5.960	kN
Vu = 1.5 X V	=	8.939	kN
Shear Stress, Tv	=	0.072	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.072	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26.000	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	36.380	
Lx/d	=	10.960	
	<	36.380	Slab Is Safe In Deflection

Slab No. :	**S4**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	5.070	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm
Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Effective Depth Along LY, Deffy	=	115.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	2-Way	
Panel Type	=	One Long Edge Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.047	
BM	=	5.186	kNm
Mu = 1.5 X BM	=	7.778	kNm

Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.062	
BM	=	6.914	kNm
Mu = 1.5 X BM	=	10.371	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.028	
BM	=	3.064	kNm
Mu = 1.5 X BM	=	4.596	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.037	
BM	=	4.085	kNm
Mu = 1.5 X BM	=	6.127	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.468	
V = SF1 x TL x Lx	=	13.301	kN
Vu = 1.5 X V	=	19.951	kN
Shear Stress, Tv	=	0.160	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.160	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.360	
V = SF2 x TL x Lx	=	10.231	kN
Vu = 1.5 X V	=	15.347	kN
Shear Stress, Tv	=	0.133	N/sqmm
Nominal Shear Stress, Tc	=	0.297	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.386	
	>	0.133	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	36.380	
Lx/d	=	31.360	
	<	36.380	Slab Is Safe In Deflection

Slab No. :	**S5**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	5.070	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Effective Depth Along LY, Deffy	=	115.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	2-Way	
Panel Type	=	Interior Panel	

Design Moments:

Short Span Positive Moment At Midspan -			
Moment Coefficient	=	0.034	
BM	=	3.839	kNm
Mu = 1.5 X BM	=	5.759	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m
Short Span Negative Moment At Continuous Support -			
Moment Coefficient	=	0.046	
BM	=	5.119	kNm
Mu = 1.5 X BM	=	7.679	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m
Long Span Positive Moment At Midspan -			
Moment Coefficient	=	0.024	
BM	=	2.674	kNm
Mu = 1.5 X BM	=	4.011	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m
Long Span Negative Moment At Continuous Support -			
Moment Coefficient	=	0.032	
BM	=	3.565	kNm
Mu = 1.5 X BM	=	5.348	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.409	
V = SF1 x TL x Lx	=	11.615	kN
Vu = 1.5 X V	=	17.422	kN
Shear Stress, Tv	=	0.139	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.139	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.330	
V = SF2 x TL x Lx	=	9.379	kN
Vu = 1.5 X V	=	14.068	kN
Shear Stress, Tv	=	0.122	N/sqmm
Nominal Shear Stress, Tc	=	0.297	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.386	
	>	0.122	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	36.380	
Lx/d	=	31.360	
	<	36.380	Slab Is Safe In Deflection

Slab No. :	**S6**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	5.070	m
Short Span, Lx	=	1.370	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Dead Load,DL	=	5.250	kN/sqm
Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	1-Way	
End Conditions	=	End Span	
Load Combination	=	1.5 DL + 1.5 LL	

Design Moments:

Short Span Positive Moment at Mid Span -

Moment Coefficient(DL), M1	=	0.083	
Moment Coefficient(LL), M2	=	0.100	
BM = M1 x DL x Lx x Lx + M2 x LL x Lx x Lx	=	1.197	kNm
Mu = 1.5 X BM	=	1.795	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Positive Reinforcement -

Area Of Reinforcement Required	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment at Continuous Support -

Moment Coefficient(DL), M3	=	0.100	
Moment Coefficient(LL), M4	=	0.111	
BM = M3 x DL x Lx x Lx + M4 x LL x Lx x Lx	=	1.402	kNm
Mu = 1.5 X BM	=	2.104	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

All Other Sections provide Minimum Steel @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Shear Coefficient(DL), SF1	=	0.400	
Shear Coefficient(LL), SF2	=	0.450	

V = SF1 x DL x Lx + SF2 x LL x Lx	=	5.960	kN
Vu = 1.5 X V	=	8.939	kN
Shear Stress, Tv	=	0.072	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.072	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26.000	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	36.380	
Lx/d	=	10.960	
	<	36.380	Slab Is Safe In Deflection

Slab No. :	**S7**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.500	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm
Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Effective Depth Along LY, Deffy	=	115.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	2-Way	
Panel Type	=	One Long Edge Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.039	
BM	=	4.370	kNm
Mu = 1.5 X BM	=	6.555	kNm

Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.052	
BM	=	5.826	kNm
Mu = 1.5 X BM	=	8.740	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.028	
BM	=	3.064	kNm
Mu = 1.5 X BM	=	4.596	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.037	
BM	=	4.085	kNm
Mu = 1.5 X BM	=	6.127	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.419	
V = SF1 x TL x Lx	=	11.913	kN
Vu = 1.5 X V	=	17.870	kN
Shear Stress, Tv	=	0.143	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.143	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.360	
V = SF2 x TL x Lx	=	10.231	kN
Vu = 1.5 X V	=	15.347	kN
Shear Stress, Tv	=	0.133	N/sqmm
Nominal Shear Stress, Tc	=	0.297	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.386	
	>	0.133	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	36.380	
Lx/d	=	31.360	
	<	36.380	Slab Is Safe In Deflection

Slab No. :	**S8**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.500	m
Short Span, Lx	=	3.920	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Effective Depth Along LY, Deffy	=	115.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	2-Way	
Panel Type	=	Interior Panel	

Design Moments:

Short Span Positive Moment At Midspan -			
Moment Coefficient	=	0.030	
BM	=	3.318	kNm
Mu = 1.5 X BM	=	4.977	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m
Short Span Negative Moment At Continuous Support -			
Moment Coefficient	=	0.040	
BM	=	4.424	kNm
Mu = 1.5 X BM	=	6.636	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m
Long Span Positive Moment At Midspan -			
Moment Coefficient	=	0.024	
BM	=	2.674	kNm
Mu = 1.5 X BM	=	4.011	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m
Long Span Negative Moment At Continuous Support -			
Moment Coefficient	=	0.032	
BM	=	3.565	kNm
Mu = 1.5 X BM	=	5.348	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.374	
V = SF1 x TL x Lx	=	10.640	kN
Vu = 1.5 X V	=	15.960	kN
Shear Stress, Tv	=	0.128	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.128	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.330	
V = SF2 x TL x Lx	=	9.379	kN
Vu = 1.5 X V	=	14.068	kN
Shear Stress, Tv	=	0.122	N/sqmm
Nominal Shear Stress, Tc	=	0.297	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.386	
	>	0.122	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	36.380	
Lx/d	=	31.360	
	<	36.380	Slab Is Safe In Deflection

Slab No. :	**S9**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	4.500	m
Short Span, Lx	=	1.370	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Dead Load,DL	=	5.250	kN/sqm
Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	1-Way	
End Conditions	=	End Span	
Load Combination	=	1.5 DL + 1.5 LL	

Design Moments:

Short Span Positive Moment at Mid Span -

Moment Coefficient(DL), M1	=	0.083	
Moment Coefficient(LL), M2	=	0.100	
BM = M1 x DL x Lx x Lx + M2 x LL x Lx x Lx	=	1.197	kNm
Mu = 1.5 X BM	=	1.795	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Positive Reinforcement -

Area Of Reinforcement Required	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment at Continuous Support -

Moment Coefficient(DL), M3	=	0.100	
Moment Coefficient(LL), M4	=	0.111	
BM = M3 x DL x Lx x Lx + M4 x LL x Lx x Lx	=	1.402	kNm
Mu = 1.5 X BM	=	2.104	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

All Other Sections provide Minimum Steel @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Shear Coefficient(DL), SF1	=	0.400	
Shear Coefficient(LL), SF2	=	0.450	

V = SF1 x DL x Lx + SF2 x LL x Lx	=	5.960	kN
Vu = 1.5 X V	=	8.939	kN
Shear Stress, Tv	=	0.072	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.072	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	26.000	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	26.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	36.380	
Lx/d	=	10.960	
	<	36.380	Slab Is Safe In Deflection

Slab No. :	**S10**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	3.920	m
Short Span, Lx	=	2.130	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm
Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Effective Depth Along LY, Deffy	=	115.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	2-Way	
Panel Type	=	Two Adjacent Edges Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.067	
BM	=	2.203	kNm
Mu = 1.5 X BM	=	3.305	kNm

Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.089	
BM	=	2.938	kNm
Mu = 1.5 X BM	=	4.406	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.034	
BM	=	1.118	kNm
Mu = 1.5 X BM	=	1.678	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.045	
BM	=	1.491	kNm
Mu = 1.5 X BM	=	2.237	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.581	
V = SF1 x TL x Lx	=	8.970	kN
Vu = 1.5 X V	=	13.455	kN
Shear Stress, Tv	=	0.108	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.108	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.400	
V = SF2 x TL x Lx	=	6.177	kN
Vu = 1.5 X V	=	9.266	kN
Shear Stress, Tv	=	0.081	N/sqmm
Nominal Shear Stress, Tc	=	0.297	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.386	
	>	0.081	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	32	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	32.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	44.775	
Lx/d	=	17.040	
	<	44.775	Slab Is Safe In Deflection

Slab No. :	**S11**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	3.920	m
Short Span, Lx	=	2.130	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Effective Depth Along LY, Deffy	=	115.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	2-Way	
Panel Type	=	One Long Edge Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.064	
BM	=	2.096	kNm
Mu = 1.5 X BM	=	3.144	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.085	
BM	=	2.795	kNm
Mu = 1.5 X BM	=	4.192	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.028	
BM	=	0.905	kNm
Mu = 1.5 X BM	=	1.357	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.037	
BM	=	1.206	kNm
Mu = 1.5 X BM	=	1.809	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.564	
V = SF1 x TL x Lx	=	8.717	kN
Vu = 1.5 X V	=	13.075	kN
Shear Stress, Tv	=	0.105	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.105	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.360	
V = SF2 x TL x Lx	=	5.559	kN
Vu = 1.5 X V	=	8.339	kN
Shear Stress, Tv	=	0.073	N/sqmm
Nominal Shear Stress, Tc	=	0.297	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.386	
	>	0.073	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	32	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	32.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	44.775	
Lx/d	=	17.040	
	<	44.775	Slab Is Safe In Deflection

Slab No. :	**S12**		
Level	=	8.077 m	
Design Code	=	IS 456 : 2000 + IS 13920 : 2016	
Grade Of Concrete	=	M25	
Grade Of Steel	=	Fe550	
Clear Cover	=	20.000	mm
Long Span, Ly	=	2.130	m
Short Span, Lx	=	1.370	m
Imposed Load	=	1.500	kN/sqm
Live Load	=	2.000	kN/sqm

Slab Thickness	=	150.000	mm
Effective Depth Along LX, Deffx	=	125.000	mm
Effective Depth Along LY, Deffy	=	115.000	mm
Self Weight	=	3.750	kN/sqm
Total Load, TL	=	7.250	kN/sqm
Span	=	2-Way	
Panel Type	=	Two Adjacent Edges Discontinuous	

Design Moments:

Short Span Positive Moment At Midspan -

Moment Coefficient	=	0.060	
BM	=	0.818	kNm
Mu = 1.5 X BM	=	1.227	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Short Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.080	
BM	=	1.091	kNm
Mu = 1.5 X BM	=	1.636	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Long Span Positive Moment At Midspan -

Moment Coefficient	=	0.034	
BM	=	0.463	kNm
Mu = 1.5 X BM	=	0.694	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Reinforcement Provided	=	T8 @ 275 C/C	
	=	183.000	sqmm/m

Long Span Negative Moment At Continuous Support -

Moment Coefficient	=	0.045	
BM	=	0.617	kNm
Mu = 1.5 X BM	=	0.925	kNm
Area Of Reinforcement Required (Mu)	=	180.000	sqmm/m
Additional Area Of Reinforcement Required	=	90.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Distribution Reinforcement @ 0.12% -

Area Of Reinforcement Required	=	150.000	sqmm/m
Reinforcement Provided	=	T8 @ 300 C/C	
	=	168.000	sqmm/m

Shear Check :

Along Short Span

Shear Coefficient(Lx), SF1	=	0.547	
V = SF1 x TL x Lx	=	5.429	kN
Vu = 1.5 X V	=	8.143	kN
Shear Stress, Tv	=	0.065	N/sqmm
Nominal Shear Stress, Tc	=	0.286	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.371	
	>	0.065	Slab Is Safe In Shear

Along Long Span

Shear Coefficient(Ly), SF2	=	0.400	
V = SF2 x TL x Lx	=	3.973	kN
Vu = 1.5 X V	=	5.960	kN
Shear Stress, Tv	=	0.052	N/sqmm
Nominal Shear Stress, Tc	=	0.297	N/sqmm
Modification Factor 'K'	=	1.300	
K x Tc	=	0.386	
	>	0.052	Slab Is Safe In Shear

Deflection Check:

Basic Span To Depth Ratio	=	32	
Span Correction Factor	=	1.000	
Tensile Steel Factor	=	1.399	
Compression Steel Factor	=	1.000	
Flange Correction Factor	=	1.000	
Max Span To Depth Ratio	=	32.00 x 1.00 x 1.40 x 1.00 x 1.00	
	=	44.775	
Lx/d	=	10.960	
	<	44.775	Slab Is Safe In Deflection